American Options by Longstaff-Schwartz: Regression-Based Optimal Stopping with a Binomial-Tree Validator

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Abstract

The American option pricing problem is fundamentally different from its European counterpart: the holder can exercise at any time up to maturity, so the price is the value function of a continuous-time optimal stopping problem. No closed form is available except in degenerate cases (zero rate, infinite maturity). This block presents the Longstaff-Schwartz (2001) regression-based Monte Carlo algorithm, which discretises the exercise dates to a Bermudan grid and replaces the conditional expectation in the dynamic-programming Bellman equation by a cross-sectional regression of continuation values onto a finite basis of polynomials of the underlying. We use Laguerre polynomials of degree up to three for the regression basis. The algorithm is validated against an independent Cox-Ross-Rubinstein binomial tree implementation, also from scratch, which serves as the ground truth in the limit of small step size.

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1 Optimal stopping and the American option

1.1 The American put

An *American put* on a single non-dividend-paying asset S with strike K and maturity T pays its holder $\max(K - S_\tau, 0)$ at any chosen time $\tau \in [0, T]$, at the holder's discretion. Under the risk-neutral measure $\mathbb{Q}$ with constant rate r, the time-zero price is

$$V_0 \;=\; \sup_{\tau \in \mathcal{T}_{[0,T]}} \mathbb{E}^{\mathbb{Q}}\big[e^{-r\tau}\,(K - S_\tau)^+\big]\,, \tag{1}$$

where $\mathcal{T}_{[0,T]}$ is the set of stopping times of the natural filtration $\mathcal{F} = \{\mathcal{F}_t\}$ taking values in $[0, T]$, and $x^+ \stackrel{\text{def}}{=} \max(x, 0)$.

The supremum is attained at an *optimal stopping time* $\tau^\star$. The map $t \mapsto S_t^\star$ separating the continuation region $\{S > S_t^\star\}$ from the exercise region $\{S \le S_t^\star\}$ is the *optimal exercise boundary*; it is a non-trivial function of t for the American put and admits no closed form.

1.2 Why no Black-Scholes formula

For the European put, the holder is committed to exercising at T, so the option price is just the discounted expectation $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(K - S_T)^+]$, which integrates against the lognormal density of S_T to produce the Black-Scholes formula. The American case introduces an entirely different kind of object: the supremum over τ in (1) is a sup over an infinite-dimensional set of measurable strategies. There is no elementary integral that captures it.

The classical machinery for handling (1) is the *Snell envelope* construction.

1.3 The Snell envelope

For a discounted reward process $X_t = e^{-rt}(K - S_t)^+$ adapted to $\mathcal{F}$, the Snell envelope U is defined as the smallest supermartingale dominating X:

$$U_t \;=\; \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}^{\mathbb{Q}}[X_\tau \mid \mathcal{F}_t]. \tag{2}$$

Under standard regularity, U_t exists and is the unique solution to the recursive equation

$$U_t \;=\; \max\big(X_t,\, \mathbb{E}^{\mathbb{Q}}[U_{t+\mathrm{d}t} \mid \mathcal{F}_t]\,\big), \qquad U_T = X_T. \tag{3}$$

The price of the American option is $V_0 = U_0$, and the optimal stopping time is $\tau^\star = \inf\{t \ge 0 : U_t = X_t\}$ — the first time the envelope touches the immediate-exercise reward.

Equation (3) is a backward dynamic programming recursion. In continuous time it leads to a free-boundary PDE (the Black-Scholes PDE with an obstacle); in discrete time, which we adopt below, it gives a clean numerical scheme.

2 Bermudan approximation and the Bellman equation

2.1 Discretising the exercise dates

A *Bermudan* option is the discrete analogue: exercisable only at a finite set of pre-specified dates $0 = t_0 < t_1 < \cdots < t_N = T$. We take an equispaced grid $t_k = kT/N$ for some large N. As $N \to \infty$, the Bermudan price converges to the American price monotonically from below [4, Section 8.3].

Let $S_k \stackrel{\text{def}}{=} S_{t_k}$. The Bermudan put value at exercise date t_k, given $S_k = s$, satisfies the Bellman recursion:

$$\boxed{\begin{aligned} V_N(s) &= (K-s)^+, \\ V_k(s) &= \max\Big(\underbrace{(K-s)^+}_{\text{intrinsic}},\ \underbrace{e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[V_{k+1}(S_{k+1}) \mid S_k = s]}_{\text{continuation value } C_k(s)}\Big), \end{aligned}} \tag{4}$$

for $k = N-1, N-2, \ldots, 0$, with $\Delta t = T/N$. Then $V_0(S_0)$ is the Bermudan put price.

The Markov property of GBM under $\mathbb{Q}$ guarantees that the conditional expectation $\mathbb{E}^{\mathbb{Q}}[V_{k+1}(S_{k+1}) \mid \mathcal{F}_k]$ depends on $\mathcal{F}_k$ only through S_k, so the continuation value $C_k(s)$ is a function of one scalar variable. This is essential for the regression-based approach below.

2.2 The optimal exercise rule

Equation (4) also gives the optimal exercise rule: exercise at the first k such that

$$(K - S_k)^+ \ \geq\ C_k(S_k).$$

At each path, the rule depends only on the comparison between the known intrinsic value and the unknown continuation value. The whole challenge is computing $C_k(s)$.

2.3 Why pure Monte Carlo fails directly

A naive attempt to estimate $C_k(s)$ would require, at each path i and each time k, simulating a sub-tree of paths from $(t_k, S_k^{(i)})$ to (t_{k+1}, S_{k+1}) and averaging the continuation values $V_{k+1}(S_{k+1})$. This is a nested simulation of cost $O(n^2 N)$ for n paths and N time steps — prohibitive for any realistic N and n. Worse, V_{k+1} is itself defined by nested simulation, so the cost compounds geometrically with k.

Longstaff and Schwartz [1] solved this in a single line of insight: *the conditional expectation is a function of S_k, so we can fit it by cross-sectional regression across the existing paths.*

3 The Longstaff-Schwartz algorithm

3.1 The regression idea

At time t_k, the unknown continuation value $C_k(\cdot) : \mathbb{R}_+ \to \mathbb{R}$ is approximated by a linear combination of M basis functions $\psi_1, \psi_2, \ldots, \psi_M$ of the underlying:

$$C_k(s) \ \approx\ \widehat{C}_k(s) \ \stackrel{\text{def}}{=}\ \sum_{m=1}^{M} \beta_m^{(k)}\, \psi_m(s). \tag{5}$$

The coefficients $\beta^{(k)} = (\beta_1^{(k)}, \ldots, \beta_M^{(k)})$ are chosen by ordinary least squares against the simulated cash flows of the optimal strategy from t_{k+1} onwards.

Concretely: simulate n independent paths $(S_0, S_1^{(i)}, \ldots, S_N^{(i)})_{i=1,\ldots,n}$ of the GBM under $\mathbb{Q}$. Process the time grid *backwards*. At step k, we have already determined, for each path i, the discounted cash flow $Y^{(i)}$ that the optimal strategy from time t_{k+1} onwards produces, evaluated at time t_k (i.e., discounted back to t_k). Regress these $\{Y^{(i)}\}_i$ on the M-vector of basis values $\{\psi_m(S_k^{(i)})\}_m$:

$$\beta^{(k)} \;=\; (\Psi_k^\top \Psi_k)^{-1} \Psi_k^\top Y, \tag{6}$$

where Ψ_k is the $n \times M$ design matrix with entries $(\Psi_k)_{i,m} = \psi_m(S_k^{(i)})$. The regression produces an estimated continuation value $\widehat{C}_k(S_k^{(i)})$ at each path.

Then update the cash flow array: at any path i where the intrinsic value $(K - S_k^{(i)})^+$ exceeds $\widehat{C}_k(S_k^{(i)})$, the optimal action is to exercise now, so $Y^{(i)}$ is replaced by $(K - S_k^{(i)})^+$. Otherwise $Y^{(i)}$ is discounted back one step: $Y^{(i)} \to e^{-r\Delta t} Y^{(i)}$.

After the recursion reaches $k = 0$, the price estimator is the sample mean of the $\{Y^{(i)}\}_i$ discounted to t_0:

$$\widehat{V}_0 \;=\; \frac{1}{n} \sum_{i=1}^{n} Y^{(i)}. \tag{7}$$

3.2 The in-the-money restriction

A critical detail of the original LS paper: the regression (6) should be performed only on the subset of paths that are *in the money* at t_k, i.e., for which $K - S_k^{(i)} > 0$. Out-of-the-money paths contribute zero intrinsic value, so the exercise vs. continuation decision is moot at those paths (always continue). Including them in the regression dilutes the signal: it forces the polynomial fit to behave reasonably where it doesn't matter, at the cost of accuracy where it does matter.

This is one of those small algorithmic choices that makes a large practical difference. We follow it in the implementation.

3.3 Algorithm pseudocode

Algorithm 1 Longstaff-Schwartz for the American put

Require: Spot S_0, params (K, r, σ, T), exercise grid N, paths n, basis $\{\psi_m\}_{m=1}^M$.

```
Simulate matrix S ∈ R^{n×(N+1)} of GBM paths.
Initialise cash flow Y^(i) ← (K − S_N^(i))^+ for all i.
for k = N − 1 downto 1 do
    Discount cash flows: Y^(i) ← e^{−rΔt} Y^(i).
    Find ITM paths: I_k = {i : K − S_k^(i) > 0}.
    Build design Ψ_k ∈ R^{|I_k|×M}, target Y_{I_k}.
    Solve β^(k) = (Ψ_k^⊤ Ψ_k)^{−1} Ψ_k^⊤ Y_{I_k}.
    Compute Ĉ_k^(i) = Σ_m β_m^(k) ψ_m(S_k^(i)) for i ∈ I_k.
    Determine exercise set E_k = {i ∈ I_k : (K − S_k^(i)) ≥ Ĉ_k^(i)}.
    For i ∈ E_k: replace Y^(i) ← K − S_k^(i).
end for
Final discount: Y^(i) ← e^{−rΔt} Y^(i) for all i.
return V̂_0 = (1/n) Σ_i Y^(i) and standard MC half-width.
```

Note on time $k = 0$. The algorithm exits the loop at $k = 1$ and applies one final discount. This is because S_0 is deterministic, so the exercise-or-continue decision at t_0 is just a comparison of the immediate intrinsic value $(K - S_0)^+$ against the sample mean $\widehat{V}_0$ produced by the

algorithm. We return the maximum of the two. (For the canonical at-the-money put this is moot, since $(K - S_0)^+ = 0$.)

4 The Laguerre basis

4.1 Definition

The Laguerre polynomials $\{L_n\}_{n\geq 0}$ are the orthogonal polynomials on $[0, \infty)$ with respect to the weight $w(x) = e^{-x}$. They are defined by the recurrence

$$L_0(x) = 1, \quad L_1(x) = 1 - x, \quad (n+1)L_{n+1}(x) = (2n + 1 - x)L_n(x) - n\, L_{n-1}(x). \tag{8}$$

Closed forms for the first few:

$$L_0(x) = 1, \tag{9}$$
$$L_1(x) = 1 - x, \tag{10}$$
$$L_2(x) = \tfrac{1}{2}(x^2 - 4x + 2), \tag{11}$$
$$L_3(x) = \tfrac{1}{6}(-x^3 + 9x^2 - 18x + 6). \tag{12}$$

Their orthogonality is $\int_0^\infty L_n(x)L_m(x)\, e^{-x}\, \mathrm{d}x \;=\; \delta_{nm}$.

4.2 Use in LSM

In our LSM implementation we apply the basis to the normalised underlying $x = S/K$, with $M = 4$ basis functions:

$$\psi_m(S) = L_{m-1}(S/K), \qquad m = 1, 2, 3, 4. \tag{13}$$

The normalisation S/K keeps the argument in a controlled range (typically in $[0, 1]$ for ITM put paths, since ITM means $S < K$). On $[0, 1]$, the unweighted Laguerre polynomials are well-behaved and linearly independent; the design matrix $\Psi_k^\top \Psi_k$ is positive-definite and Cholesky-factorisable.

Why Laguerre and not monomials? The original Longstaff-Schwartz paper uses weighted Laguerre $\tilde{\psi}_m(x) = e^{-x/2}L_{m-1}(x)$, which is orthonormal under $\mathrm{d}x$. The weight makes the basis bounded uniformly on $[0, \infty)$. Our use of the unweighted form is acceptable because we restrict regression to ITM paths (so $x \in [0, 1]$ effectively), where the weight is close to constant. The numerical behaviour is essentially identical; we choose unweighted for code simplicity. Plain monomials $\{1, x, x^2, x^3\}$ also work and are commonly used by practitioners; the choice of basis matters less than the principle of *using one*.

5 The Cox-Ross-Rubinstein binomial tree validator

The LSM estimator is statistically biased and depends on basis choice, sample size, and number of exercise dates. To validate it we need an independent reference price computed by a different method. The standard choice is the Cox-Ross-Rubinstein (CRR) *binomial tree* [2], which converges to the true American price as the tree depth increases.

5.1 Construction

Discretise time into N steps of size $\Delta t = T/N$. At each step the asset goes up by factor u or down by factor d, with risk-neutral probabilities $(p, 1-p)$:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u = e^{-\sigma\sqrt{\Delta t}}, \quad p = \frac{e^{r\Delta t} - d}{u - d}. \tag{14}$$

The choice $u \cdot d = 1$ keeps the tree recombining: after j ups and $k - j$ downs, the asset price is $S_0 \cdot u^{2j-k}$. The total number of nodes after N steps is $(N+1)(N+2)/2$, so the tree fits in memory for N up to several thousand.

5.2 Backward induction

At maturity ($k = N$), the option value at each terminal node is the intrinsic put value. At earlier nodes, we apply the Bellman equation exactly (no regression needed):

$$V_k(j) \;=\; \max\bigl(K - S_k(j),\; e^{-r\Delta t}[\,p\,V_{k+1}(j+1) + (1-p)\,V_{k+1}(j)\,]\bigr), \tag{15}$$

where $V_k(j)$ is the option value at node j of step k and $S_k(j) = S_0 u^{2j-k}$. The price is $V_0(0)$.

5.3 Convergence

The CRR price converges to the true American price as $N \to \infty$ at rate roughly $1/\sqrt{N}$ (see [5, Section 4.5]); in practice $N = 1000$ to 5000 is sufficient for four-decimal accuracy on a one-year ATM American put with $\sigma = 0.20$.

6 Bias and convergence of the LSM estimator

6.1 Two sources of error

The LSM estimator $\widehat{V}_0^{\mathrm{LSM}}$ has two distinct sources of error compared to the true Bermudan price:

1. **Stochastic error**: the Monte Carlo sampling uncertainty, of order $1/\sqrt{n}$ in the number of paths. This is the standard MC error and is captured by the asymptotic half-width.
2. **Approximation bias**: the regression-based continuation value $\widehat{C}_k$ is an approximation of the true C_k. Two effects compete:
   - *Sub-optimal stopping*: any approximation of C_k leads to a stopping rule different from $\tau^\star$, which gives a strictly lower expected payoff (because $\tau^\star$ is by definition the supremum). This is a **negative bias** (LSM tends to underprice).
   - *In-sample bias*: when we use the same paths to estimate $\widehat{C}_k$ and to apply the resulting rule, the rule is data-mined to look better than it really is on those paths. This is a **positive bias** (LSM tends to overprice).

In practice the two biases approximately cancel: the LSM estimator is near-unbiased for sufficiently many paths and a sufficiently rich basis. The literature documents agreement with reference values to better than 1% on the canonical American put problem with $n = 10^5$ and $M = 3$ to 4 Laguerre basis functions.

6.2 The unbiased upper bound

A rigorously unbiased *lower bound* estimator of the price can be obtained by splitting the simulation into two independent batches: the first batch is used to fit the regression coefficients, the second to apply the resulting exercise rule and average the cash flows. This eliminates the in-sample bias at the cost of a constant factor of paths.

A separate *upper bound* can be constructed via the Andersen-Broadie (2004) duality approach, which exploits the martingale formulation of the optimal stopping problem. We do not implement either refinement in this block; we rely on the near-cancellation of the two biases and validate against the binomial tree.

7 Empirical validation strategy

The validation script implements four tests.

Test 1: binomial tree convergence. Compute the binomial American put at $N = 100, 500, 1000, 5000, 1000$ The sequence should converge to a stable value at four decimal places by $N = 5000$.

Test 2: LSM matches binomial. At canonical parameters ($S = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$), with $N_{LSM} = 50$ exercise dates and $n = 10^5$ paths, the LSM estimator should agree with the binomial reference (at $N_{tree} = 5000$) within three half-widths.

Test 3: early exercise premium. Compute both the European put (Black-Scholes) and the American put (LSM, plus binomial as sanity). The American minus European spread is the early-exercise premium; it must be strictly positive, and large enough to be distinguishable from MC noise (the canonical ATM case gives a premium of order 0.1 on a put of order 5.6).

Test 4: input validation. Mechanical.

8 Discussion: when LSM works and when it does not

LSM is the workhorse method for high-dimensional American options where binomial and PDE methods are infeasible. Its strength is that the cost scales linearly in the dimension of the underlying state, in contrast to the exponential blowup of grid-based methods. For our single-asset American put it is overkill — the binomial tree is faster, more accurate, and entirely deterministic — but the value of the implementation is in the methodology: LSM is the standard tool for path-dependent and multi-asset Americans (basket Americans, swing options, real options on multi-factor commodities).

When LSM fails or struggles. Three regimes:

1. *Deeply out-of-the-money options*: the regression has few in-the-money paths, so the continuation function is poorly identified. Workaround: use importance sampling to bias paths into the money.

2. *Highly path-dependent payoffs*: if the option's value depends on the entire history rather than just S_k, a one-dimensional regression is insufficient. Workaround: include path-summary statistics in the basis (the running maximum, average, etc.).

3. *Discontinuous payoffs*: digital Americans have discontinuous intrinsic values, which destabilise the regression near the discontinuity. Workaround: smoothing via known regularisations.

For the smooth, single-state American put we treat in this block, none of these difficulties arise.

9 Implementation outline

The block ships two pricers in a new module `quantlib.american` (and the C++ mirror):

- `binomial_american_put`: CRR validator. No randomness, just the lattice recursion. Used as the ground truth.
- `lsm_american_put`: Longstaff-Schwartz with Laguerre basis $M = 4$, in-the-money restriction, exact GBM stepping.

The C++ side adds a file-local helper `cholesky_solve` that factorises the small $M \times M$ normal-equations matrix and back-substitutes for $\beta^{(k)}$. For our $M = 4$ basis, this is a direct rather than iterative solve; the matrix is small, positive-definite, and well-conditioned in practice.

10 Bibliographic notes

The Longstaff-Schwartz algorithm is from [1]. Glasserman [4, Chapter 8] gives a careful textbook treatment with full convergence theory. The Cox-Ross-Rubinstein binomial tree is from [2]; modern textbook treatment in Hull [5, Chapter 21]. Andersen and Broadie's upper-bound construction is in [3].

The treatment of optimal stopping via Snell envelopes is classical; see e.g. Karatzas and Shreve [6] for a rigorous measure-theoretic exposition.

References

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